

operations summary

Here's a **crisp, exam-focused summary** of your notes so you can revise quickly:

Operations / Production Management – Summary

1. What is Operations Management?

- Planning, designing, and operating **production systems** to achieve organizational goals
 - Applies to **manufacturing + service systems**
 - Goal is **not just cost minimization**, but:
 - Profitability
 - Productivity
 - Long-term demand
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2. Evolution

- Industrial Revolution → Scientific Management
 - WWII → Operations Research (OR), Management Science
 - Modern → Automation, IT systems
 - Japanese System → **Quality first + continuous improvement + waste elimination**
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3. Types of Production Systems

- **Unit / Job / Batch / Mass / Continuous**
 - Increasing order:
 - 👉 Flexibility ↓, Volume ↑, Automation ↑
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4. Operations Strategy

- Must align with **overall business strategy**
- Impacts:

- Cost
- Quality
- Delivery
- Flexibility

Key Strategies

- Capacity strategy
 - Quality (TQM)
 - Technology
 - Location
 - Product flexibility
 - Quick delivery
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5. Productivity

- **Productivity = Output / Input**
 - Measures:
 - Output per labor
 - Value added
 - Must consider **long-term impact**, not just short-term gains
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6. Performance Measures

- Quality
 - On-time delivery
 - Safety
 - Employee turnover
 - Machine breakdown
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7. Lean Manufacturing (Very Important)

Core Idea

👉 Eliminate waste + continuous improvement

7 Wastes

- Overproduction
 - Waiting
 - Inventory
 - Motion
 - Transport
 - Defects
 - Overprocessing
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Key Tools

- **JIT (Just in Time)** → produce only when needed
 - **Kaizen** → continuous improvement
 - **5S** → workplace organization
 - **Kanban** → visual inventory control
 - **Poka-Yoke** → mistake proofing
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8. JIT (Just-In-Time)

- Produce based on **actual demand**
 - Features:
 - Low inventory
 - Small batch sizes
 - Faster response
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9. Total Quality Management (TQM)

- Customer-focused
 - Everyone responsible for quality
 - Continuous improvement
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10. Business Process Reengineering (BPR)

- **Radical redesign** of processes
 - Goal:
 - Reduce cost
 - Improve speed
 - Increase efficiency
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11. World Class Manufacturing (WCM)

Based on

- Lean
- TQM
- TPM (maintenance)

Key Pillars

- Safety
 - Quality
 - Cost
 - Maintenance
 - Logistics
 - People development
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12. Maintenance (TPM)

- Prevent breakdowns
 - Train operators for basic maintenance
 - Aim → **maximum equipment efficiency**
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13. Quality & Value Concepts

- **Value Engineering** → reduce cost without affecting function
 - **Taguchi Method** → robust design
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14. EHSS (Safety & Environment)

- Environment
 - Health
 - Safety
 - Security
 - Aim → zero incidents, sustainability
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15. Indian Industry Context

- Growing manufacturing hub
 - Strong in:
 - Automobiles
 - Chemicals
 - Pharma
 - Moving towards:
 - WCM
 - Industry 4.0
 - Automation (robots, IoT)
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FINAL SUPER SHORT REVISION (10-sec recall)

- 👉 OM = Input → Process → Output → Feedback
 - 👉 Lean = eliminate waste
 - 👉 JIT = produce when needed
 - 👉 TQM = quality everywhere
 - 👉 BPR = redesign processes
 - 👉 WCM = Lean + TQM + TPM
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If you want, I can compress this into a **1-page handwritten-style cheat sheet** (very useful before exam).